

History of the Computer



CodeHS

So... What is a
computer?

What is the
bare-bones
definition of a
computer?

When I think “computer”...



What is a Computer?

- A computer ...
 - Is a computing machine
 - Performs calculations according to precise instructions
 - It does these operations incredibly fast!

First Definition

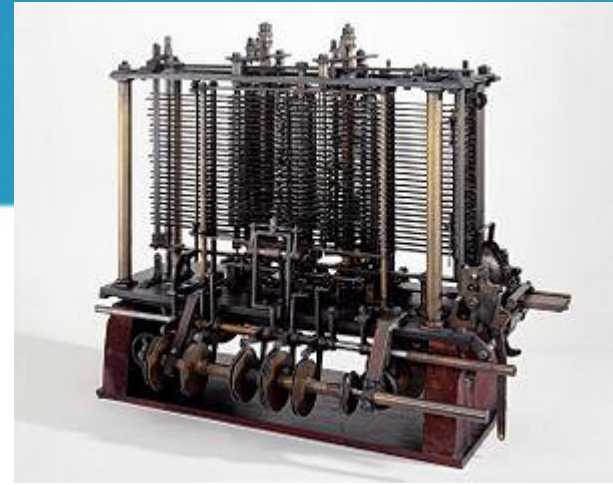
- 1613 - the word “computer” first used
 - Computer - a person who performed calculations or computations
- Computers were people!

Where did
computers
come from?

The timeline of
computer
development

First Computers

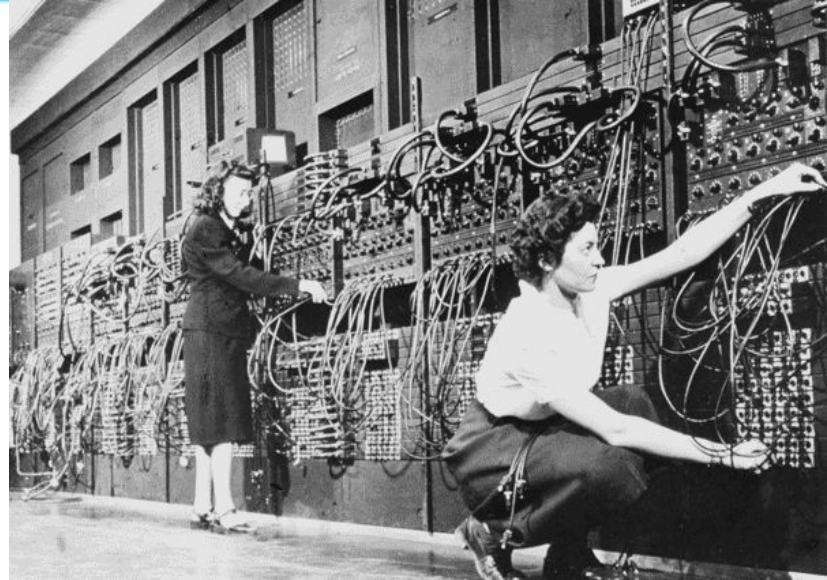
- 2600 BC: Chinese abacus
- 1910: Henry Babbage constructs part of Charles Babbage's Analytical Engine
 - Could perform basic mathematical calculations



Analytical Engine^[1]

Early Computers

- 1936: Konrad Zuse's Z1
 - First binary digital computer
- 1936: Alan Turing's Turing Machine theory
 - First universal computing machine
- 1946: ENIAC
 - First electronic general purpose computer



Women programming the ENIAC^[2]

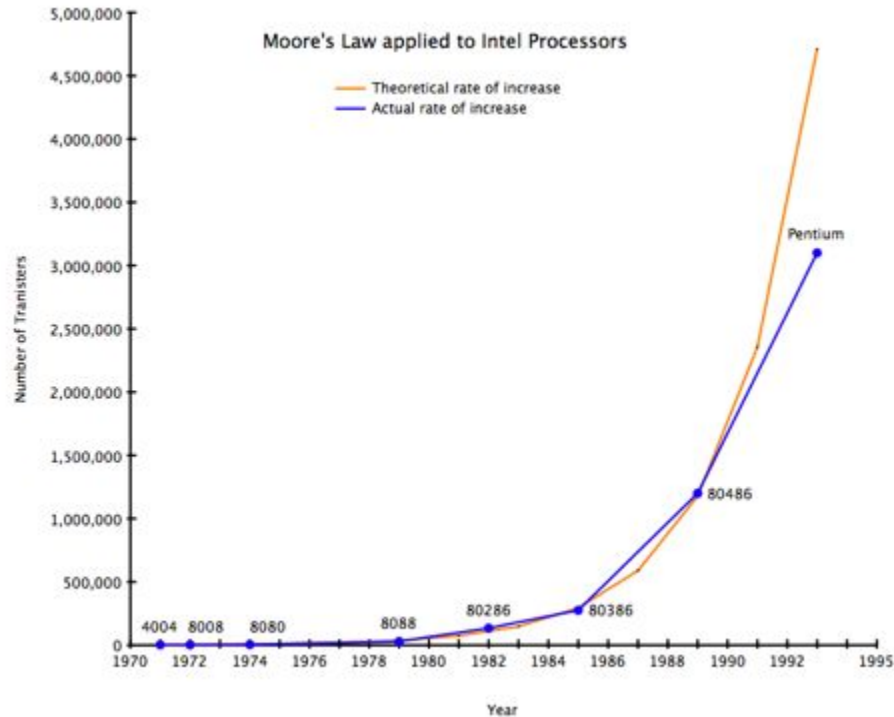
First Modern Computers

- 1950: UNIVAC
 - First modern computer
- 1975: First portable computer from IBM
 - Weighed 55 lbs with a 5 inch display and 64KB of memory



IBM's first portable computer[4]

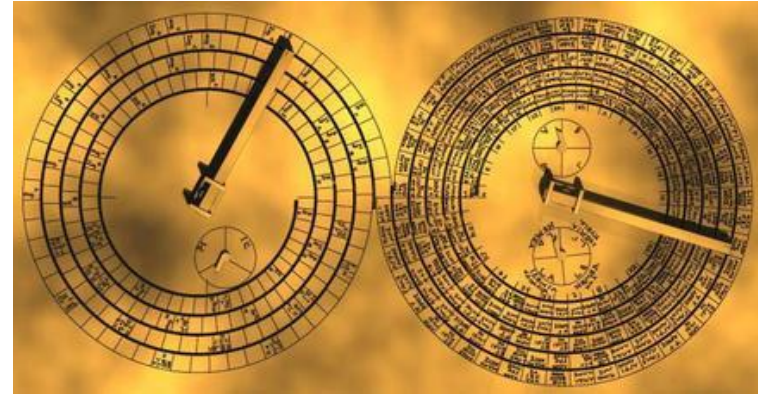
Moore's Law



Why did people
need
computers?

First Uses of Computers

- To keep counts
 - Livestock
- Marketplaces
 - Summing money
- Calculate astronomical positions
 - Navigation
 - Calendars



Computer recreation of back panel of Greek Antikythera mechanism; used for predicting astronomical positions and eclipses ^[5]

Uses Driving Development

- To solve math problems
 - Solve them faster
 - Solve more complex problems
- Military uses
 - Decode encrypted messages during wars
 - Compute proper trajectory of missiles
- Business uses
 - Automate production of goods

Modern Computers

What do computers
look like today?

What are
computers used for
today?

Modern Computers

- Laptops
- Desktops
- Tablets



Laptop, Desktop, and tablet examples^[4]

But Computers are also...

- Phones
- Smart Watches
- Toys
- Cars
- Appliances



Programmable coffee maker



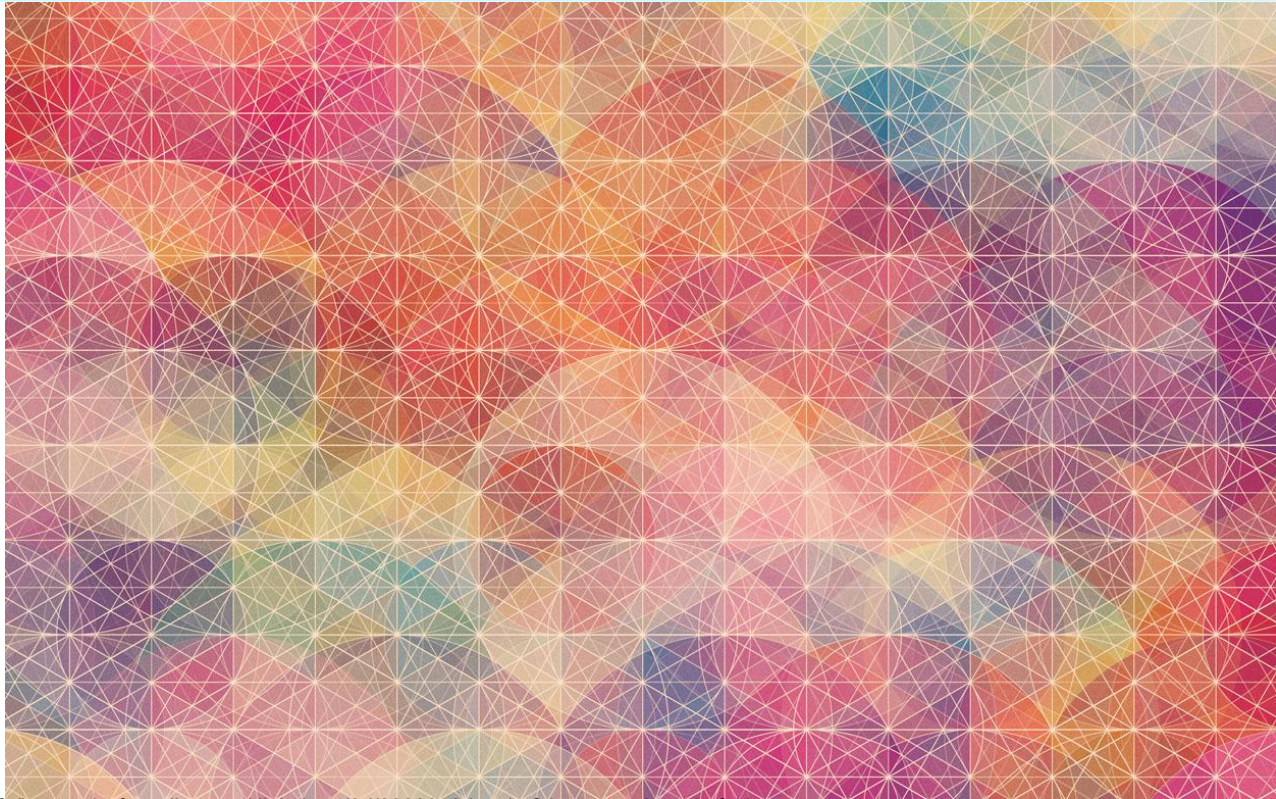
Robotic pet dog

(Some) Applications of Computers

- Auto-pilot
 - Planes
 - Cars
- Medical Devices
 - Defibrillator
 - Pacemakers

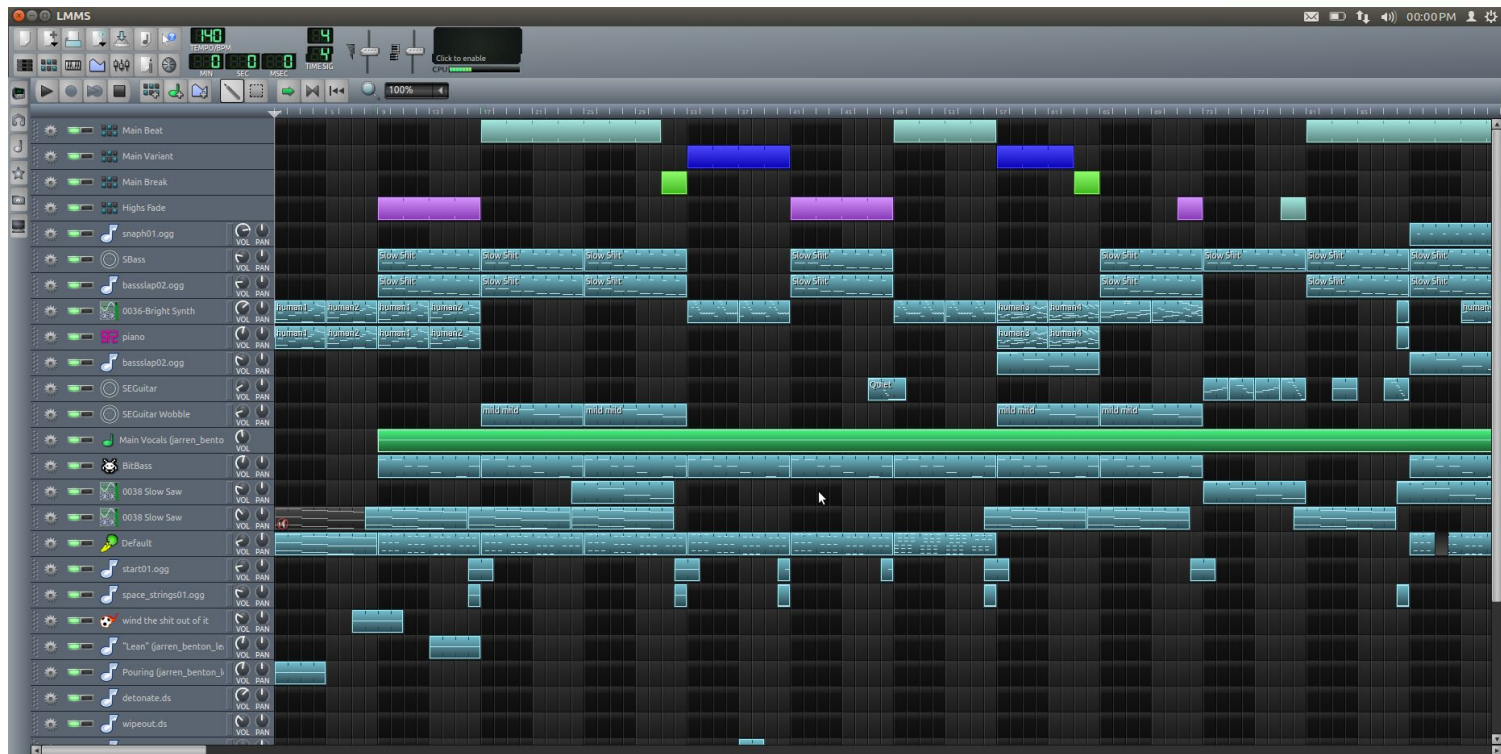


Art

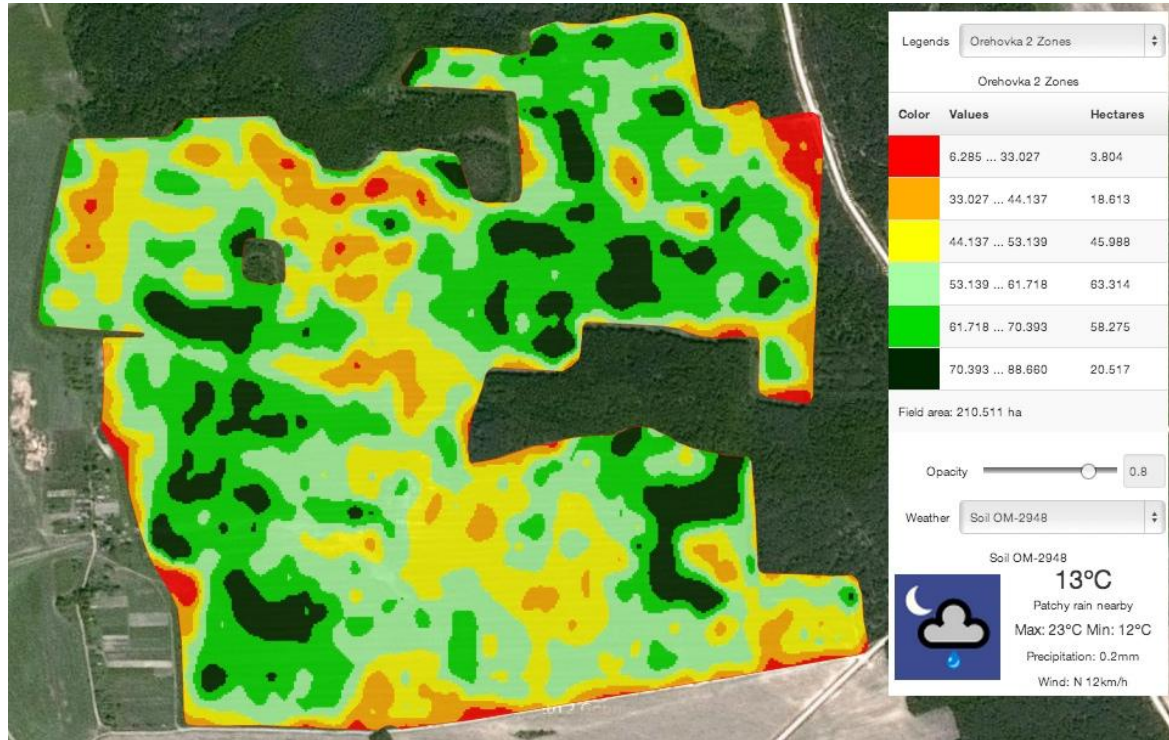


source: http://www.imgbase.info/images/safe-wallpapers/digital_art/2d/23086_2d_colorful_gemoetric_colorful.jpg

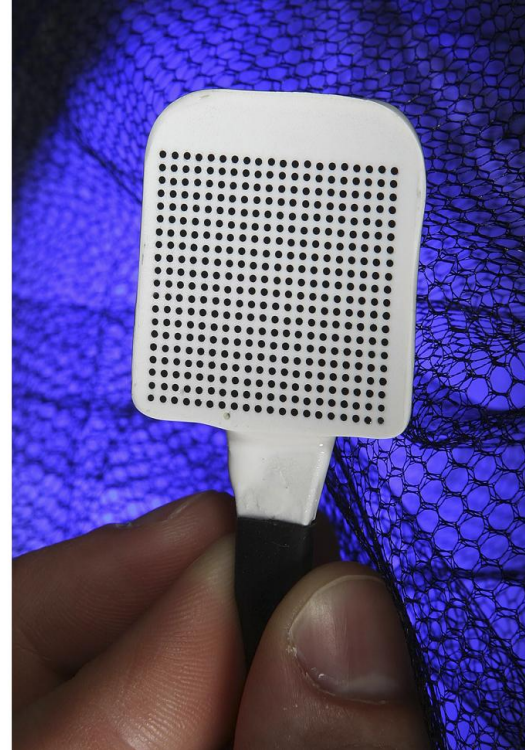
Music



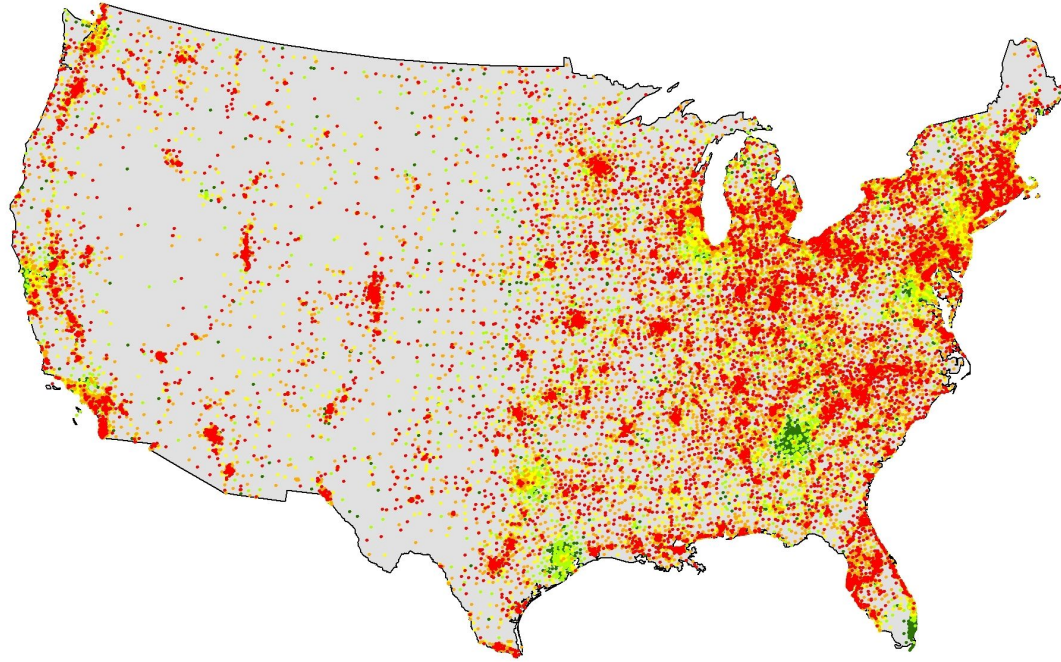
Agriculture



Health and Medicine



Health and Medicine



Illness Prevalence (%): 0.01 - 1.00 1.01 - 2.00 2.01 - 3.00 3.01 - 4.00 4.01 - 5.00

Connection



Sources

Pictures:

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- [3] By Sandstein (Own work) [CC BY-SA 3.0 (<http://creativecommons.org/licenses/by-sa/3.0>)], via Wikimedia Commons
- [4] Davis, Beth H. "The Academic Minute: Technology & Education." ProVisions Teaching and Learning Series. N.p., 14 Apr. 2016. Web. 13 July 2016.
- [5] By Source (WP:N FCC#4), Fair use, <https://en.wikipedia.org/w/index.php?curid=36445604>

Content:

- https://en.wikipedia.org/wiki/Computer#Pre-twentieth_century
- https://en.wikipedia.org/wiki/Antikythera_mechanism
- <https://plus.maths.org/content/why-was-computer-invented-when-it-was>